Table 3.4: Gram nature of common pathogenic bacteria

	Gram positive bacteria	Gram negative bacteria
Cocci	*Staphylococcus aureus, Streptococcus pyogenes*	*Neisseria gonorrhoeae*
Rods(bacilli)	*Mycobacterium tuberculosis, Bacillus anthracis, Corynebacterium diphtheriae, Clostridium tetani*	*Escherichia coli, Shigella Salmonella, Pseudomonas aeruginosa*
Spirochaetes	—	*Leptospira, Treponema*

3.8 Special Staining – Endospore Staining

Endospores are highly resistant structures produced by some bacteria during unfavourable environment conditions. Endospore formation is a distinguishing feature of aerobic genera *Bacillus* and anaerobic genera *Clostridium*. The size, shape and position of the spore (Figure 3.9) are relatively constant characteristics of a given species and are important in identifying the species within genera. The position of spore in the cell may be terminal, central or sub-terminal. Figure 3.9 shows the position of spores in a vegetative cell.

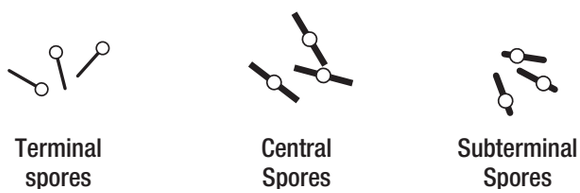


Figure 3.9: Position of spore in a vegetative cell.

Endospores cannot be stained by ordinary methods, such as simple staining and Gram staining, because the dyes do not penetrate the wall of the endospore. If simple stains are used, the vegetative body of the bacillus is deeply coloured, whereas the spore is unstained and appears as a clear area in the organism.

By vigorous staining procedure, the dye can be introduced into the spore. Once stained, the spore tends to retain the dye even after treatment with decolorizing agents. The most commonly used endospore staining procedure is the Schaeffer Fulton endospore staining method. Malachite green, the primary stain, is applied to a heat fixed smear and heated to steaming for about 5 minutes. Heat helps the stain to penetrate the endospore wall. Then the preparation is washed for about 30 seconds with water. Next safranin, a counterstain is applied to the smear to stain the portions of the cell other than endospores.

In a properly prepared smear, the endospores appear green within red cells (Figure 3.10). Endospores are highly refractive. They can be detected under the light microscope when unstained, but cannot be differentiated from inclusions of stored material without a special stain.



Figure 3.10: Schaeffer Fulton Endospore staining method- spores stained green and vegetative cell stained pink

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